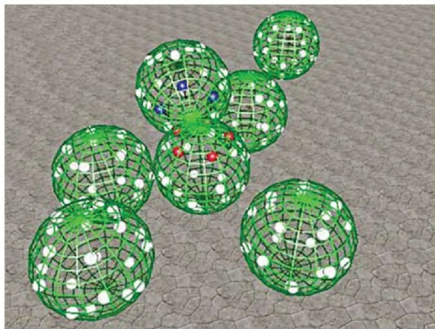


CLAYTRONICS: ENABLING TELEPORTATION

As research progresses on this exciting field, the potential to fax a synthetic replica of yourself to another location seems a possibility! This project aims to create nanoscale robotic mechanisms with computing abilities



Claytronic atoms or catoms

■ DR S.S. VERMA

Transmission and reception of sound energy by telephone was followed by a revolution in electronics and its digitalisation. Pictures along with sound were then transported successfully from one place to another along with the miniaturisation of electronics. Such developments motivated scientists and engineers to think about human teleportation. Teleportation is the transfer of matter from one place to another instantaneously, either by paranormal means or through technological artifice. Matter teleportation, however, is still a day dream, but along these lines researchers have come up with a new science

called claytronics. The idea is to create small robots of a few millimeters in size (perhaps even a few nanometers) and then have them organise themselves into a shape that is determined remotely.

An emerging field

Claytronics (also known as 'programmable matter') is an emerging field of engineering concerning reconfigurable nanoscale robots (claytronic atoms or catoms) designed to form large-scale machines or mechanisms. The catoms would be sub-millimeter computers that eventually gain the ability to move around, communicate with other computers, change colour, and electrostatically connect to other

catoms to form different shapes when instructed by the software to do so. The forms made up of catoms could morph into any object, even replicas of human beings for virtual meetings.

According to researchers at Carnegie Mellon University, US, claytronics is described as "An ensemble of material that contains sufficient local computation, actuation, storage, energy, sensing and communication," which can be programmed to form interesting dynamic shapes and configurations.

Origin of research

Claytronics research arose out of a combination of work on micro-scale computing devices and on telepresence. To get these claytronic manifestations organise themselves, it has been suggested that they adjust the size and locations of empty chambers within a group's general structure to form raised areas or troughs. This would allow their overall shape to be controlled delicately. Photo-sensors and pressure sensors would allow input to be transmitted to any location required.

Claytronics is a more workable version of nanotechnology, which in its most advanced form promises to do the same work but requires billions of self-assembling robots controlled with computerised information. Claytronic technology has the potential to become a reality because of the ever-increasing speed of computer processing predicted in Moore's Law. As the processors are getting smaller and smaller, at the sub-millimeter